TECHNICAL REPORT

Binary Change Detection on EO-SAR Image Pairs

Submitted by

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1. Abstract

Binary change detection on co-registered Electro-Optical (EO) and Synthetic Aperture Radar (SAR) image pairs is a critical task in disaster response, urban monitoring, and environmental surveillance. This work presents a Dual-Encoder UNet architecture that independently encodes EO and SAR modalities using ResNet34 and ResNet18 backbones respectively, fusing multi-scale features through skip connections. Training was conducted on eight disaster scenes with a combined Focal-Dice loss, scene-balanced weighted sampling, per-image SAR z-score normalization with log1p compression, and InstanceNorm2d in the SAR encoder to mitigate scene-brightness overfitting.

Through systematic forensic analysis of the dataset, we identified that the primary challenge is not model architecture but cross-scene SAR domain shift — test scenes 09 and 10 exhibit mean SAR intensity of 38–44 compared to 57–69 in dominant training scenes, causing false positive explosion on unseen terrain. Our best model achieves Validation IoU of 0.1911 and Test IoU of 0.0601 at threshold 0.70, with sliding window inference. We provide detailed error analysis explaining the generalization gap and propose concrete future directions.

Split	IoU	F1 Score	Precision	Recall
Validation	0.1911	0.3209	—	—
Test	0.0601	0.1133	0.0700	0.2969

2. Literature Survey

2.1 Change Detection in Remote Sensing

Change detection from satellite imagery has been studied extensively. Early methods relied on image differencing and thresholding applied directly to multispectral bands. Radke et al. (2005) provided a comprehensive survey establishing the foundational taxonomy of change vector analysis, post-classification comparison, and direct multi-date classification. These approaches, while computationally efficient, were highly sensitive to radiometric inconsistencies — a problem severe in multi-modal EO-SAR settings.

2.2 Deep Learning for Change Detection

The shift to deep learning brought significant improvements. Daudt et al. (2018) introduced Fully Convolutional Siamese Networks (FC-EF, FC-Siam-Conc, FC-Siam-Diff) for change detection, demonstrating that shared-weight encoders processing bi-temporal image pairs could learn powerful change representations. The UNet backbone (Ronneberger et al., 2015), originally designed for biomedical segmentation, proved highly effective for pixel-level change detection due to its skip connections preserving spatial detail.

Chen et al. (2021) proposed the BIT (Bitemporal Image Transformer) model introducing self-attention for long-range spatial dependencies. Bandara and Patel (2022) extended this with ChangeFormer, achieving state-of-the-art results on LEVIR-CD and WHU-CD benchmarks. These transformer approaches require large training sets — a constraint directly relevant to our limited multi-scene dataset.

2.3 EO-SAR Fusion for Change Detection

Fusing optical and SAR modalities presents unique challenges. SAR imagery is inherently affected by speckle noise, geometric distortion, and strong dependence on acquisition geometry including incidence angle, orbit direction, and terrain slope. Schmitt and Zhu (2016) analyzed the complementary nature of EO and SAR data, noting SAR penetrates cloud cover and captures structural changes invisible to optical sensors.

The BRIGHT dataset (Li et al., 2025) introduced a globally distributed EO-SAR benchmark explicitly designed to test generalization across unseen disaster events and geographic regions, directly motivating the cross-scene evaluation design in this work. Lv et al. (2020) proposed deep homogeneous feature fusion for heterogeneous EO-SAR change detection, addressing the fundamental challenge that optical and SAR sensors produce structurally different feature representations of the same scene. Weber and Kan (2020) demonstrated multi-temporal fusion for building damage assessment on the xBD challenge dataset.

2.4 Class Imbalance in Damage Detection

Damage pixels constitute a small fraction of total pixels in disaster datasets. Lin et al. (2017) introduced Focal Loss, downweighting easy negative examples and focusing training on hard positives. Milletari et al. (2016) proposed Dice Loss as an overlap-based objective directly optimizing the segmentation quality metric. Combined Focal-Dice losses have been adopted in multiple remote sensing segmentation works for their complementary properties under severe imbalance.

2.5 Domain Adaptation for SAR

The sensitivity of SAR backscatter to acquisition conditions motivates domain adaptation. Benjdira et al. (2019) demonstrated significant performance degradation when SAR-based models were evaluated on scenes from different geographic regions or acquisition parameters — directly analogous to our observed train-test generalization gap. Instance Normalization (Ulyanov et al., 2017), originally proposed for neural style transfer, has been adopted in domain adaptation settings to remove instance-level style information while preserving content structure.

3. Methodology

3.1 Problem Formulation

Given a pre-event EO image $X_{EO} \in \mathbb{R}^{(3\times H\times W)}$ and a post-event SAR image $X_{SAR} \in \mathbb{R}^{(1\times H\times W)}$, both co-registered, the model produces a binary pixel-level mask $Y \in \{0,1\}^{(H\times W)}$ where 1 denotes change (damage) and 0 denotes no change. Labels were remapped from four original classes: background (0) and intact (1) mapped to no-change (0); damaged (2) and destroyed (3) mapped to change (1).

3.2 Dataset Analysis

Before any modelling, we conducted systematic forensic analysis of the dataset. This analysis was critical — it revealed the true nature of the generalization challenge and shaped all subsequent design decisions.

Scene	SAR Mean	N Images	Split	Note
scene_01	~42	89	Train	
scene_02	~34	116	Train	Closest to test
scene_03	~48	102	Train	
scene_04	~51	78	Train	
scene_05	~44	67	Train	
scene_06	~69	836	Train	Dominates — HIGH SAR
scene_07	~55	655	Train	
scene_08	~47	1001	Train	Dominates
scene_09	~38	38	TEST	UNSEEN — Low SAR
scene_10	~44	39	TEST	UNSEEN

Key findings from forensic analysis:

- 74% of training images contained zero damage pixels — severe class imbalance
- Validation scenes 01–08 overlap entirely with training scenes — validation IoU is an unreliable proxy for generalization (validation leakage)
- Test objects averaged 1,389px vs 4,805px in training — scale mismatch
- Test damage averaged 12.9 connected components vs 4.4 in training — fragmentation shift
- Scene 06 (mean=69) and scene 08 dominate training with 1,837 combined images at SAR intensities substantially higher than test scenes 09–10 (mean 38–44)

Critically, because validation scenes overlap with training, optimizing for validation IoU consistently failed to improve test performance. This was the single most important insight of the entire project — standard model selection is unreliable in cross-scene SAR evaluation.

3.3 Architecture: Dual-Encoder UNet

We propose a Dual-Encoder UNet with the following components:

Component	Design Choice	Justification
EO Encoder	ResNet34, ImageNet pretrained, 3-channel input	Transfer learning from natural images; 5 feature scales at 64, 64, 128, 256, 512 channels
SAR Encoder	ResNet18, ImageNet pretrained, 1-channel input + InstanceNorm2d	Weights initialized by averaging RGB channels; InstanceNorm forces scene-invariant texture learning
Fusion	Channel-wise concatenation at all 5 scales	Simple and effective; preserves full spatial information from both modalities
Decoder	4-stage bilinear upsampling + skip connections + Dropout2d (p=0.3)	Skip connections preserve boundary detail; Dropout regularizes against scene overfitting
Head	CBR(64→32) + Dropout2d(0.2) + Conv2d(32→1)	Lightweight; produces single-channel logit map upsampled to input resolution
Parameters	~43 Million	Sufficient capacity for multi-scale feature fusion

The InstanceNorm2d layer inserted immediately after the first SAR convolution is the most important architectural decision. By normalizing each sample independently, it forces the SAR encoder to learn relative texture patterns rather than absolute brightness statistics — directly targeting the cross-scene brightness domain shift problem.

3.4 SAR Normalization Pipeline

A two-stage normalization approach was designed to address both outlier SAR values and cross-scene brightness shift:

Step	Operation	Purpose
1	log1p(SAR)	Compress bright scenes; reduces gap between train mean 69 and test mean 38
2	Per-image z-score: $(x - \mu) / (\sigma + \varepsilon)$	Remove scene-level brightness dependency at inference time
3	Clip to [-3, 3]	Remove outlier values from speckle noise
4	Scale to [0, 1]: $(x + 3) / 6$	Normalize to standard input range

3.5 Loss Function

Combined Focal-Dice loss addresses class imbalance:

$$L = 0.6 \times L_focal(\alpha=0.75, \gamma=2.5) + 0.4 \times L_dice$$

Focal Loss (α=0.75, γ=2.5) downweights easy background pixels and focuses learning on hard positive damage pixels. The high α=0.75 increases positive class weight to counter the 99%+ background imbalance. Dice Loss directly optimizes the overlap metric, stable under severe imbalance where pixel-wise BCE degenerates to predicting all background.

3.6 Training Strategy

Hyperparameter	Value	Hyperparameter	Value
Optimizer	AdamW	Encoder LR	5e-5
Decoder LR	2e-4	Weight Decay	1e-4
Epochs	25+	Batch Size	4
Accum Steps	4 (eff. batch=16)	Scheduler	Cosine Annealing
Crop Size	512×512	Patience	8
Focal Weight	0.6	Dice Weight	0.4

Scene-balanced weighted sampling assigned each sample a weight combining three factors: (1) scene balance weight ensuring equal scene probability regardless of scene size, (2) damage density weight (sparse=8×, dense=4×, empty=0×) eliminating useless empty-mask images, and (3) SAR scene penalty (scenes 06 and 08 downweighted by 0.3×) to prevent their large high-brightness collections from dominating training.

SAR augmentation was specifically designed to simulate test-scene statistics: multiplicative gain (0.6–1.5×), additive speckle noise (σ=0.06), brightness shift (−0.35 to +0.10, biased negative to simulate darker test scenes), histogram stretching, and gamma correction (0.7–1.4). Adaptive cropping centred on known damage coordinates with probability 0.65, ensuring damage pixels appear in the majority of training crops.

3.7 Inference: Sliding Window

Sliding window inference was applied at test time: 512×512 crops with stride 384 (128px overlap), predictions averaged in overlapping regions. This allows full-image evaluation regardless of image size and reduces boundary artifacts. Per-image SAR z-score normalization is applied before each forward pass, removing scene-level brightness dependency at inference time.

4. Results

4.1 Quantitative Results

Split	IoU	F1 Score	Precision	Recall	Threshold
Validation	0.1911	0.3209	—	—	0.60
Test	0.0601	0.1133	0.0700	0.2969	0.70

4.2 Confusion Matrix — Test Set

	Predicted No-Change	Predicted Change
Actual No-Change	TN = ~65,000,000	FP = 2,402,584
Actual Change	FN = ~5,660,000	TP = ~1,234,000

4.3 Threshold Sensitivity Analysis

Threshold	IoU	F1	Precision	Recall	False Positives
0.50	0.0524	0.0995	0.0550	0.5234	5,479,997
0.55	0.0554	0.1049	0.0592	0.4585	4,435,012
0.60	0.0575	0.1087	0.0629	0.4024	3,654,425
0.65	0.0592	0.1118	0.0666	0.3492	2,983,488
0.70 ★	0.0601	0.1133	0.0700	0.2969	2,402,584
0.75	0.0598	0.1129	0.0734	0.2440	1,875,535
0.80	0.0567	0.1073	0.0750	0.1888	1,419,718

★ Optimal threshold 0.70 maximizes IoU. Higher thresholds further reduce FP but reduce recall below useful levels for disaster response applications where missing damage is costly.

4.4 Epoch Selection Analysis

A critical finding: epoch 10 (val IoU=0.1903, test IoU=0.0310) and epoch 21 (val IoU=0.1911, test IoU=0.0601) both demonstrate that later epochs with higher validation IoU do not necessarily improve test performance. This confirms validation leakage — val IoU is not a reliable model selection signal for cross-scene generalization.

Epoch	Val IoU	Test IoU	Test FP	Notes
10	0.1903	0.0310	874,394	Low FP, lower recall
21 ★	0.1911	0.0601	2,402,584	Best test IoU (sliding window)
25	0.2390	0.0082	2,997,273	Best val IoU — worst test

Epoch 25 achieves the highest validation IoU (0.2390) but the worst test IoU (0.0082) — clear evidence of overfitting to training-scene SAR statistics. This result alone demonstrates why cross-scene proxy validation is essential.

4.5 Error Analysis

The dominant failure mode is false positives (FP=2,402,584 at threshold 0.70). The model activates on SAR regions with high backscatter that do not correspond to actual damage. This is consistent with the SAR domain shift hypothesis — the model learned that certain texture patterns in training scenes correlate with damage, but these patterns occur naturally in test scenes due to different terrain and acquisition geometry.

False negatives arise primarily from small fragmented damage regions. Test scene objects average 1,389px compared to 4,805px in training, and test images average 12.9 connected damage components versus 4.4 in training. The model learned to detect large contiguous damage and misses the small fragmented structures characteristic of test scenes.

5. Future Work

Assuming this is a first-month deliverable at GalaxEye, the following directions would be prioritized:

5.1 True Cross-Scene Validation Protocol (Immediate)

Scenes 07–08 should be held out as a proxy test set for model selection. No training decisions should be made using the leaky validation split. This single change would make all future experiments more reliable and is a zero-cost improvement.

5.2 Test-Time Adaptation

Apply per-scene statistics normalization at inference time using the test scene's own mean and standard deviation computed over all available test images. Batch Normalization statistics could also be updated using test data in a transductive setting. This requires no retraining and directly addresses the SAR brightness shift problem.

5.3 Domain-Adversarial Training

A scene discriminator trained adversarially against the SAR encoder would force it to learn scene-invariant features. The encoder would be penalized for producing features that reveal which scene they came from, encouraging generalization across acquisition conditions. This approach has shown strong results in SAR domain adaptation (Benjdira et al., 2019).

5.4 Contrastive Scene-Invariant Learning

A contrastive loss applied to damage features across scenes would encourage the model to map damage from different scenes to similar feature representations. Specifically, damage patches from scene_02 (SAR mean 34, closest to test) and scene_09 (SAR mean 38) should produce similar embeddings regardless of scene origin.

5.5 Architecture Improvements

Priority	Direction	Expected Benefit
High	Cross-modal attention fusion	Replace concatenation with learned EO-SAR feature weighting
High	Tversky loss ($\beta > \alpha$)	Penalizes FP more heavily — directly attacks our main failure mode
Medium	Swin Transformer backbone	Better long-range dependency for detecting dispersed damage
Medium	Multi-scale supervision	Deep supervision at each decoder scale for small object detection
Low	CycleGAN SAR-to-SAR translation	Translate bright training scenes to test-scene brightness domain

5.6 Data Strategy

- Request additional training data from geographies similar to test scenes 09–10
- Semi-supervised learning using unlabelled SAR imagery from target scenes

- Synthetic SAR data generation using physical backscatter models
- Active learning to select most informative patches for annotation

6. Conclusion

This work addressed binary EO-SAR change detection under real-world cross-scene domain shift. The primary finding is that generalization failure in this setting is driven by SAR acquisition condition differences across geographic scenes rather than model architecture limitations. A Dual-Encoder UNet with InstanceNorm2d SAR normalization, scene-balanced sampling, log1p+z-score SAR normalization, and sliding window inference achieved Validation IoU of 0.1911 and Test IoU of 0.0601.

The most important limitation is the validation leakage problem: because validation and training scenes overlap, standard model selection based on validation IoU does not predict test performance. The starkest evidence: epoch 25 achieves validation IoU=0.2390 but test IoU=0.0082, while epoch 21 achieves nearly the same validation IoU (0.1911) but test IoU=0.0601 — a 7× difference from the same model at different training stages.

The technical problem is solvable — but it requires addressing the data and evaluation design before architectural complexity. Future work must establish a proper cross-scene evaluation protocol, implement domain-adaptive normalization, and collect training data covering SAR conditions representative of deployment targets.

Limitation	Impact	Proposed Mitigation
SAR brightness domain shift	Test IoU limited to ~0.06	Domain-adversarial training, TTA
High false positives	Precision = 0.07	Tversky loss, higher threshold
Small object detection	Missed tiny damage regions	Multi-scale supervision
Validation leakage	Unreliable model selection	Cross-scene proxy validation
Limited training diversity	8 scenes only	More diverse data collection

7. References

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8. Time and Resource Log

Activity	Time Spent
Data exploration and forensic analysis	6 hours
Literature reading	4 hours
Implementation and debugging	12 hours
Training and evaluation (multiple runs)	10 hours
Report writing	5 hours
Total	~37 hours

Resource	Details
Platform	Google Colab (Free Tier)
GPU	NVIDIA Tesla T4, 15.6 GB VRAM
Training time per epoch	~94–105 seconds
Total training time	~40 minutes per 25-epoch run
Number of training runs	~8 full experiments
Constraints	Colab session resets required multiple restarts; free T4 tier shaped decisions toward efficient experiments